

Quiz 2

1. C/D definition

Write the mathematical relationship between a continuous-time signal $x_c(t)$ and its samples $x[n]$. Define T and f_s .

$x[n] = x_c(nT)$, $T = 1/f_s$ is sampling period [sec/sample], $f_s = 1/T$ is sampling rate [samples/sec]

2. Sampled Spectrum

For ideal impulse-train sampling, write $X_s(j\Omega)$ in terms of $X_c(j\Omega)$, T , and Ω_s . What physical picture does this equation describe?

$X_s(j\Omega) = \frac{1}{T} \sum_k X_c(j(\Omega - k\Omega_s))$, replication of $X_c(j\Omega)$ at multiples of $\Omega_s = \frac{2\pi}{T}$ and scaling by $\frac{1}{T}$.

3. Alias calculation

A 9 kHz sinusoid is sampled at 12 kHz. What sinusoidal frequency appears in the sampled data when interpreted in [0,6] kHz?

3 kHz

4. Nyquist condition

A signal is strictly bandlimited to 4.5 kHz. What is its ideal Nyquist rate? Is $f_s = 8$ kHz sufficient?

Nyquist rate is $2f_M = 9$ kHz here. $f_s = 8$ kHz is less than 9 kHz and not sufficient.

5. D/A concept

What frequency-domain magnitude effect is introduced by a zero-order hold?

sinc-shaped high frequency droop (-3.92 dB at $f_s/2$) and time delay by $T/2$.

6. Quantization

Estimate the ideal full-scale sine SQNR in dB for a 10-bit uniform quantizer.

$10 \cdot 6.02 + 1.76 = 61.96$ dB